Lesson 2.1 - How to Make a Simple Frequency Table

> **Video:** How To Make a Simple Frequency Table by The Organic Chemistry Tutor **Chapter:** 2 - Graphs and Tables

What is a frequency table?

A **frequency table** has two columns:

V (value)	f (frequency)
each different value in the data	how many times that value occurs

How to build one:

1. Find the **lowest** value and count how many times it appears.
2. Move to the **next** value up and count again.
3. Repeat until you reach the highest value. Skip values that do not appear.

Using a frequency table to find the mean, median, mode, and range

Measure	How to read it from the table
Range	highest value − lowest value
Mode	the value with the **largest frequency** (two tied → **bimodal**)
Median	the middle value; count in from both ends using the frequencies
Mean	$\bar{x} = \frac{\sum(V \cdot f)}{\sum f}$; multiply each value by its frequency, add, and divide by $n = \sum f$

Example 1

The data contains these values and counts:

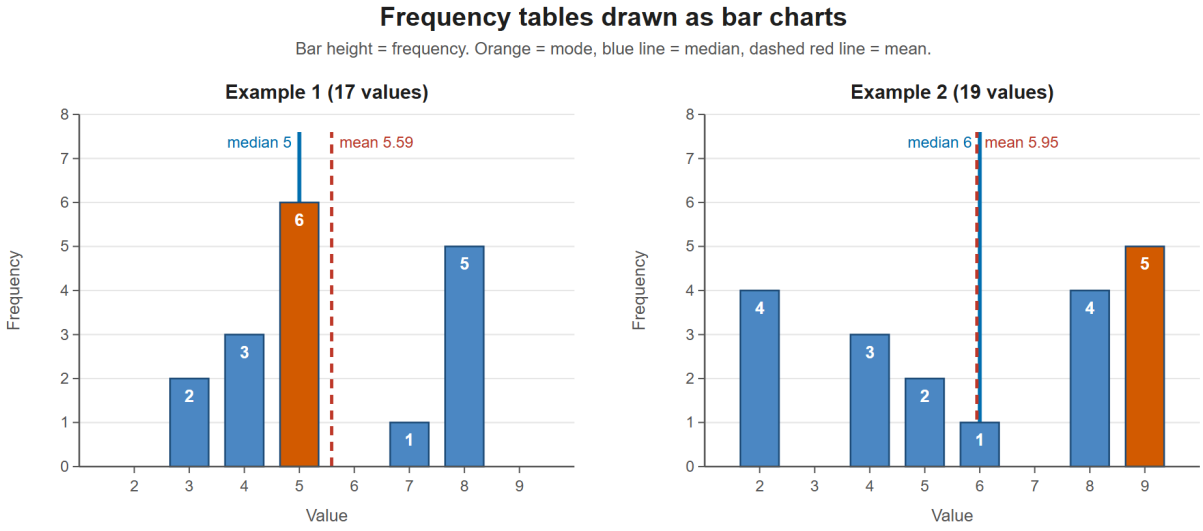
V	f	V · f
3	2	6
4	3	12
5	5	25
7	1	7
8	5	40

- **Range:** $8 - 3 = 5$
- **Mode:** 5 and 8 both have the highest frequency (5) → **bimodal**, modes 5 and 8.

Now add one more 5. The frequency of 5 becomes **6**:

V	f	V · f
3	2	6
4	3	12
5	**6**	**30**
7	1	7
8	5	40
Total	**17**	**95**

- **Range:** still $\max - \min = 8 - 3 = 5$.
- **Mode:** now only **5** has the highest frequency → **unimodal**, mode = 5.
- **Median:** $n = 17$, so the median is the 9th value. Counting up: positions 1-2 are 3s, 3-5 are 4s, 6-11 are 5s. The 9th value is **5**.
  - Check by writing it out: 3 3 4 4 4 5 5 5 **5** 5 5 7 8 8 8 8 8
- **Mean:** $\bar{x} = \dfrac{\sum xf}{\sum f} = \dfrac{95}{17} \approx 5.59$



Each frequency table drawn as a bar chart. The tallest bar (orange) is the mode, the solid blue line is the median and the dashed red line is the mean. In Example 1 the mode and the median are both 5. In Example 2 the mode is 9 while the median is 6, which shows that the three measures of center can disagree.

Example 2 (practice)

V	f	V · f
2	4	8
4	3	12
5	2	10
6	1	6
8	4	32
9	5	45
Total	**19**	**113**

- **Range:** $\max - \min = 9 - 2 = 7$
- **Mode:** 9 (frequency 5, the largest)
- **Median:** $n = 19$, so the median is the 10th value. Running counts: 4 (2s), 7 (4s), 9 (5s), **10** (6). The 10th value is **6**.
  - Written out: 2 2 2 2 4 4 4 5 5 **6** 8 8 8 8 9 9 9 9 9
- **Mean:** $\bar{x} = \dfrac{\sum xf}{\sum f} = \dfrac{113}{19} \approx 5.95$

Key takeaways

- A frequency table lists each **value** and its **frequency**.
- n = the **sum of the frequency column**.
- Total of all data = $\sum(V \cdot f)$, so **mean** = $\sum(V \cdot f)/n$.
- **Mode** = value with the highest frequency. **Range** = highest − lowest.
- **Median** = the value at position $(n+1)/2$ when n is odd, or the average of the two middle values when n is even. Use running totals of f to find it.

Lesson 2.2 - Relative Frequency Distribution Table

> **Video:** How To Make a Relative Frequency Distribution Table by The Organic Chemistry Tutor **Chapter:** 2 - Graphs and Tables

Idea

Relative frequency is the **fraction** (or percentage) of the data taken up by each value:

$$\text{relative frequency} = \frac{f}{\sum f} = \frac{f}{n}$$

Steps

1. Make a frequency table: **V** (value) and **f** (frequency).
2. Add the frequency column to get n.
3. Add a third column **RF** = f/n.
4. **Check:** the RF column must add up to **1** (or 100%).

Example

Data (20 values): 2, 5, 3, 4, 7, 2, 5, 8, 3, 2, 2, 3, 2, 5, 5, 4, 8, 5, 2, 8

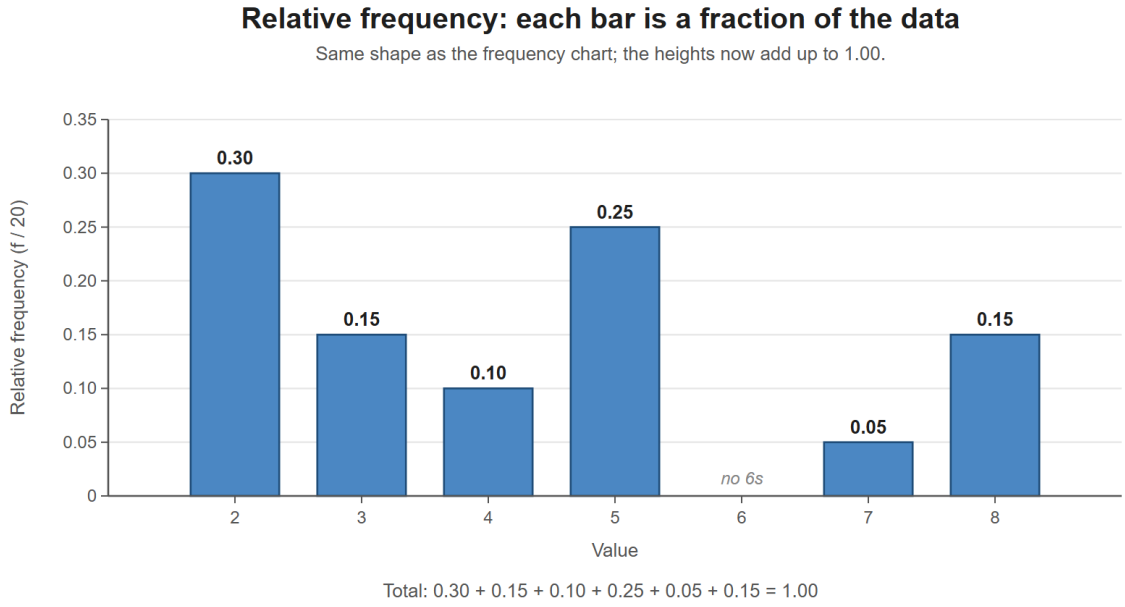
V	f	RF = f / 20
2	6	0.30
3	3	0.15
4	2	0.10
5	5	0.25
7	1	0.05
8	3	0.15
Total	**20**	**1.00**

There is no 6 in the data, so 6 is left out of the table.

Reading the table:

- The relative frequency of the value **4** is **0.10**, so 10% of the data are 4s.
- The relative frequency of the value **8** is **0.15**, so 15% of the data are 8s.

Check: $0.30 + 0.15 + 0.10 + 0.25 + 0.05 + 0.15 = 1.00$ ✓



The relative frequency chart has the same shape as a frequency chart; only the vertical scale changes. Every bar is a fraction of the 20 values, so the heights add up to exactly 1.00.

Key takeaways

- RF = frequency ÷ total frequency.
- RF values are between 0 and 1 and always **sum to 1**. If they don't, recount.
- Multiply by 100 to read RF as a percentage.
- The next step is the **cumulative relative frequency** (Lesson 2.3).

Lesson 2.3 - Cumulative Relative Frequency Table

> **Video:** How To Make a Cumulative Relative Frequency Table by The Organic Chemistry Tutor **Chapter:** 2 - Graphs and Tables

Idea

Cumulative relative frequency (CRF) is a **running total** of the relative frequencies. For each value, it gives the fraction of the data that is **less than or equal to** that value.

$$\mathrm{CRF}_k = \mathrm{RF}_1 + \mathrm{RF}_2 + \cdots + \mathrm{RF}_k$$

Steps

1. Columns: **V** (value), **f** (frequency), **RF** (relative frequency), **CRF** (cumulative relative frequency).
2. Fill in V and f, then find $n = \sum f$.
3. RF = f/n. Check that the RF column adds to 1.
4. CRF: start with the first RF, then keep adding the next RF.
5. **Check:** the last CRF must be **1.00**.

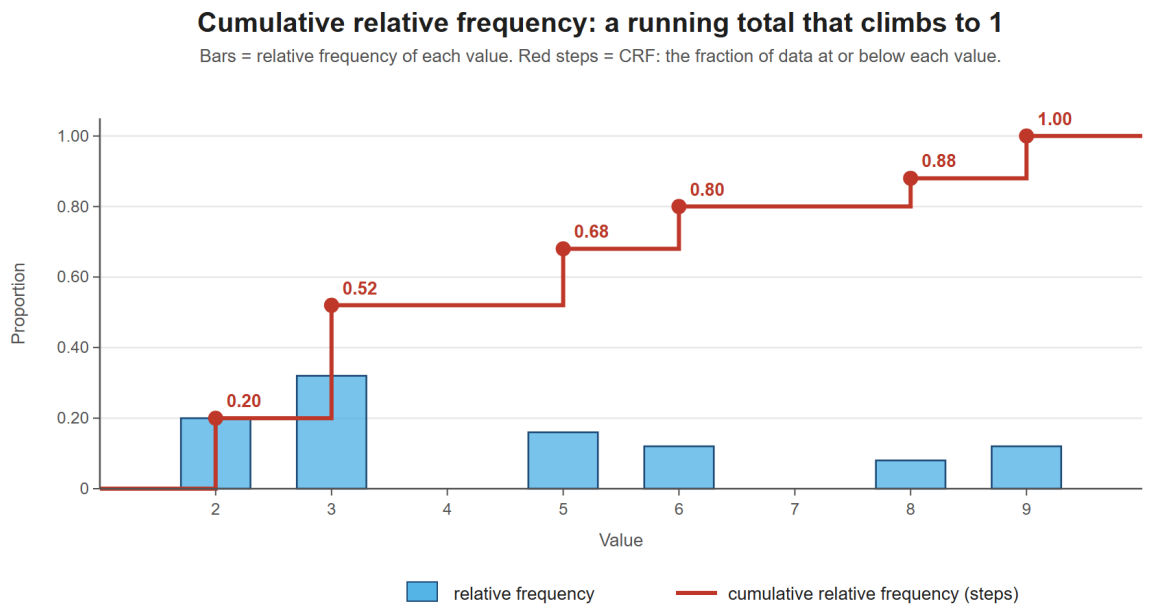
Example

The data has 25 values with these counts (there are no 4s or 7s):

V	f	RF = f / 25	CRF
2	5	0.20	0.20
3	8	0.32	0.20 + 0.32 = **0.52**
5	4	0.16	0.52 + 0.16 = **0.68**
6	3	0.12	0.68 + 0.12 = **0.80**
8	2	0.08	0.80 + 0.08 = **0.88**
9	3	0.12	0.88 + 0.12 = **1.00**
Total	**25**	**1.00**	

Reading the CRF column:

- 0.52 at V = 3 → 52% of the values are 3 or less.
- 0.80 at V = 6 → 80% of the values are 6 or less.



The cumulative relative frequency only changes at data values, so it is drawn as steps: flat until the next value, then a jump equal to that value's relative frequency. It never goes down and it ends at 1.00. Read it as "the fraction of the data at or below this value" (e.g. 0.80 of the values are 6 or less).

Key takeaways

- CRF = running sum of RF. It never decreases and always ends at **1**.
- CRF answers "what fraction of the data is **at or below** this value?"
- That makes the CRF column useful for **percentiles and quartiles** (see Lesson 1.1 and Chapter 4).

Lesson 2.4 - Dot Plots and Frequency Tables

> **Video:** Dot Plots and Frequency Tables by The Organic Chemistry Tutor **Chapter:** 2 - Graphs and Tables

Idea

A **dot plot** is a picture of a frequency table. Draw a number line from the lowest to the highest value, then stack **one dot per data value** above it. The height of each stack equals the frequency.

Example 1

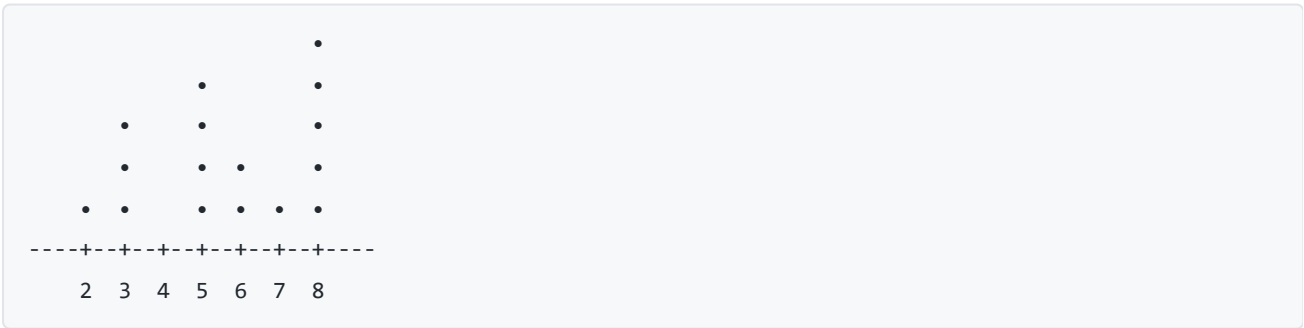
Data (16 values): 5, 8, 3, 6, 8, 5, 8, 2, 3, 8, 5, 8, 3, 5, 7, 6

Step 1 - frequency table

Value	Frequency
2	1
3	3
5	4
6	2
7	1
8	5
Total	**16**

(There is no 4.)

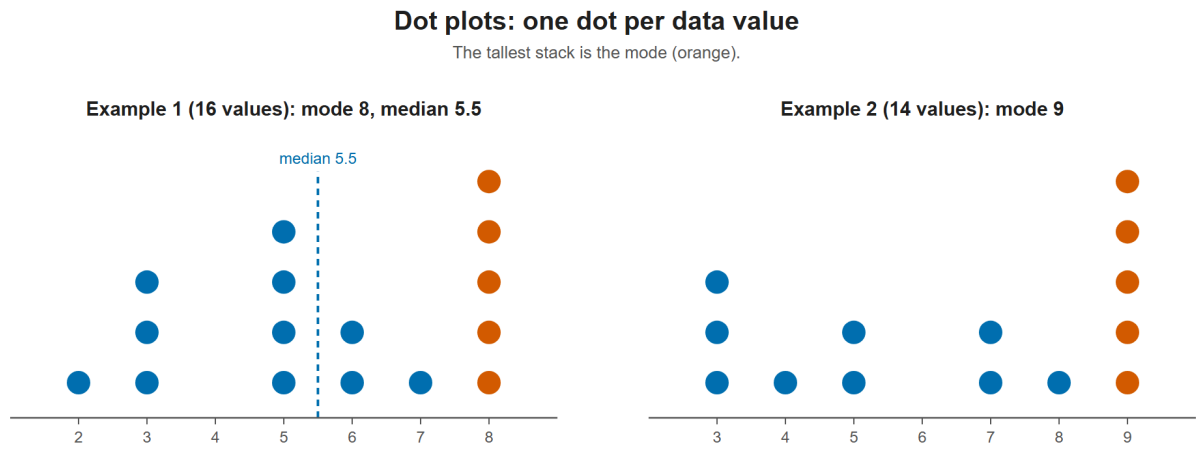
Step 2 - dot plot



Reading the plot:

- **Mode** = the tallest stack = **8**.
- **Median:** with $n = 16$, the median is the average of the 8th and 9th values. Removing dots from both ends leaves a 5 and a 6, so

$$\text{median} = \frac{5 + 6}{2} = 5.5$$



The two dot plots from this lesson. The tallest stack (orange) is the mode. In Example 1 the 8th and 9th values are a 5 and a 6, so the median is 5.5 (dashed line).

Example 2 (practice)

Data (14 values): 7, 4, 9, 3, 9, 5, 9, 7, 3, 9, 3, 9, 5, 8

Counts: 3 → 3, 4 → 1, 5 → 2, 7 → 2, 8 → 1, 9 → 5 (no 6)

```
                  •
                  •
    •             •
    •     •     •     •
    •  •  •     •  •  •
----+--+--+--+--+--+--+----
    3  4  5  6  7  8  9
```

Mode = 9 (tallest stack).

Key takeaways

- A dot plot shows **every** data value, so no information is lost.
- Number of dots above a value = its **frequency**.
- **Mode** = tallest column. For the **median**, cross off dots from both ends until the middle is left.

Lesson 2.5 - Line Plots

> **Video:** A Beginner's Guide To Line Plots by The Organic Chemistry Tutor **Chapter:** 2 - Graphs and Tables

Idea

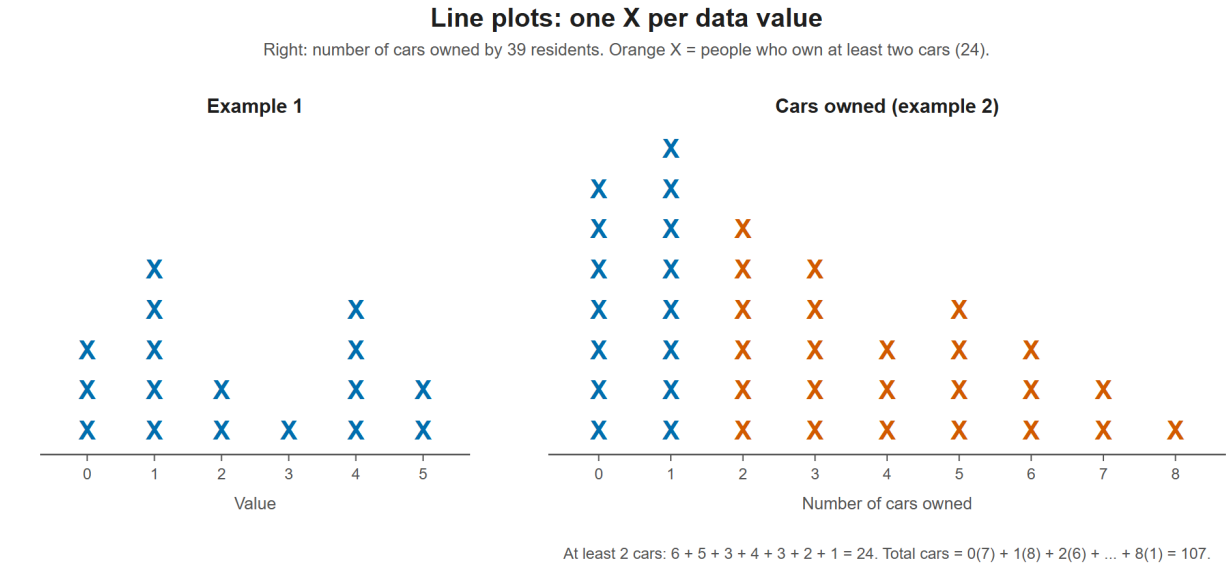
A **line plot** is the same idea as a dot plot. Draw a number line and put an **X** above a value each time it appears in the data.

Example 1 - drawing a line plot

The data contains: three 0s, five 1s, two 2s, one 3, four 4s, two 5s.

1. Sort the values: 0 0 0 1 1 1 1 1 2 2 3 4 4 4 4 5 5
2. Draw a number line from 0 to 5.
3. Stack one X per value.

```
      X
      X       X
   X  X       X
   X  X  X    X  X
   X  X  X  X X  X
----+--+--+--+--+--+--
   0  1  2  3  4  5
```



Left: the line plot from Example 1. Right: the car survey from Example 2; the orange X's are the 24 people who own at least two cars.

Example 2 - reading a line plot

A survey asked everyone in a 55-and-older community how many cars they own. The line plot shows:

Cars owned (x)	0	1	2	3	4	5	6	7	8
Number of people (X's)	7	8	6	5	3	4	3	2	1

(a) How many people own exactly one car? Count the X's above 1 → **8**.

(b) How many own *at least* two cars ($x \geq 2$)? Add everything from 2 upward:

$$\#(x \geq 2) = f_2 + f_3 + \cdots + f_8 = 6 + 5 + 3 + 4 + 3 + 2 + 1 = 24$$

(c) How many own *at most* three cars ($x \leq 3$)? Add 0 through 3, including 3:

$$\#(x \leq 3) = f_0 + f_1 + f_2 + f_3 = 7 + 8 + 6 + 5 = 26$$

(d) How many own four or five cars? $f_4 + f_5 = 3 + 4 = 7$

(e) How many people live in the community? Assuming everyone answered, add all the X's:

$$n = \sum f = 7 + 8 + 6 + 5 + 3 + 4 + 3 + 2 + 1 = 39$$

Here f_k is the number of X's above k cars.

(f) How many cars are there in total? Multiply each number of cars by how many people own that many, then add:

x	People	Cars = $x\times$ people
0	7	0
1	8	8
2	6	12
3	5	15
4	3	12
5	4	20
6	3	18
7	2	14
8	1	8
Total	**39**	**107**

There are **107 cars** in the community.

Key takeaways

- One X per data value. The height of a column is the frequency.
- **"At least k"** means $x \geq k$. **"At most k"** means $x \leq k$. Both **include** k.
- Total number of people = total number of X's.
- Total amount (cars) = $\sum(\text{value} \times \text{frequency})$, the same idea as computing a mean from a frequency table.

Lesson 2.6 - Stem and Leaf Plots

> **Video:** Stem and Leaf Plots by The Organic Chemistry Tutor **Chapter:** 2 - Graphs and Tables

Idea

A **stem-and-leaf plot** sorts the data and shows its shape while keeping every original value.

- **Stem** = every digit except the last (the "tens" part for two-digit numbers).
- **Leaf** = the last digit.
- Write the leaves in **increasing order**, with **no commas**.
- Keep a stem even if it has **no leaves**, so the gap stays visible.
- Always give a **key**, such as `2 | 4 = 24`, because the same plot could mean 24, 2.4 or 240.

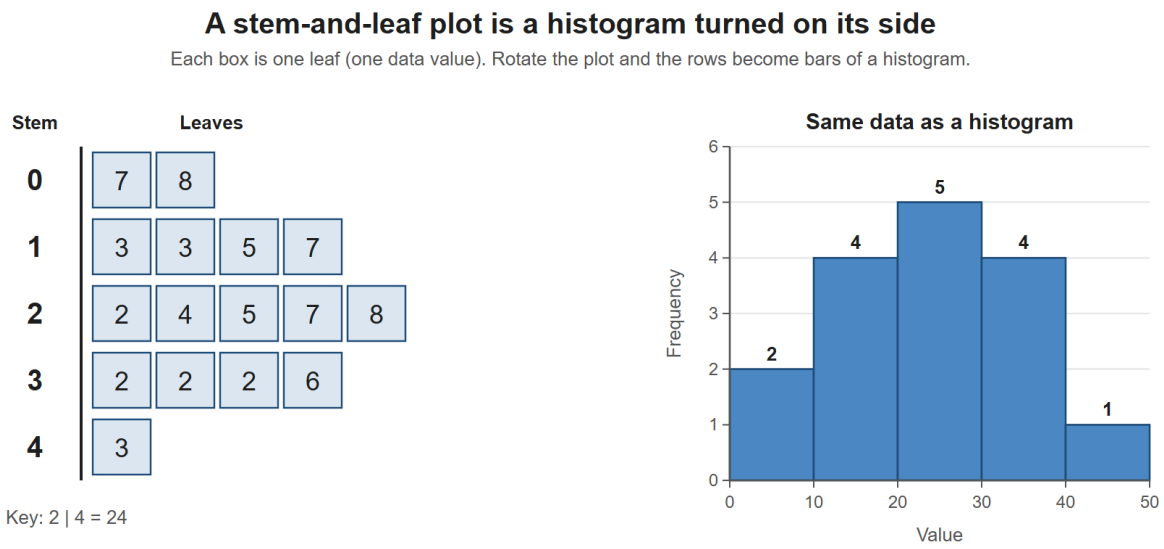
Tip: group the numbers by tens and sort them before you build the plot.

Example 1 - two-digit numbers

Sorted data: 7, 8, 13, 13, 15, 17, 22, 24, 25, 27, 28, 32, 32, 32, 36, 43

Stem	Leaf
0	7 8
1	3 3 5 7
2	2 4 5 7 8
3	2 2 2 6
4	3

Key: `1 | 3 = 13`. Single-digit numbers use a stem of 0 (`0 | 7 = 7`).



Example 1 as a stem-and-leaf plot (each box is one leaf) next to a histogram of the same 16 values with classes 0-9, 10-19, ... Turn the stem-and-leaf plot 90 degrees counter-clockwise and the rows of leaves become the bars.

Example 2 - numbers above 100

Data: 72, 76, 85, 88, 89, 93, 97, 102, 108, 113, 115

Stem	Leaf
7	2 6
8	5 8 9
9	3 7
10	2 8
11	3 5

Key: `8 | 5 = 85`, `10 | 2 = 102`. For three-digit numbers, the stem holds the first **two** digits.

Example 3 - decimals

Data: 1.2, 1.5, 1.8, 2.3, 2.4, 2.7, 2.9, 3.2, 3.6, 4.1, 4.5, 5.8, 7.6, 9.3, 10.6, 10.9, 12.4

Stem	Leaf
1	2 5 8
2	3 4 7 9
3	2 6
4	1 5
5	8
6	
7	6
8	
9	3
10	6 9
11	
12	4

Key: `1 | 2 = 1.2`, `12 | 4 = 12.4`. The empty stems 6, 8 and 11 show there are no values in those ranges.

Key takeaways

- Stem = leading digit(s), leaf = last digit, leaves sorted and without commas.

- Don't skip empty stems.
- **Always include a key**, because it tells the reader the place value.

Lesson 2.7 - Side by Side Stem and Leaf Plots

> **Video:** Side by Side Stem and Leaf Plots by The Organic Chemistry Tutor **Chapter:** 2 - Graphs and Tables

Idea

A **side-by-side** (back-to-back) stem-and-leaf plot compares **two groups** that share one stem column:

```
Group A leaves | Stem | Group B leaves
```

- **Right side:** leaves increase **left** → **right**, as usual.
- **Left side:** leaves increase **right** → **left**, going *away* from the stem. Read the left side backwards: `8 | 1` means **18**.

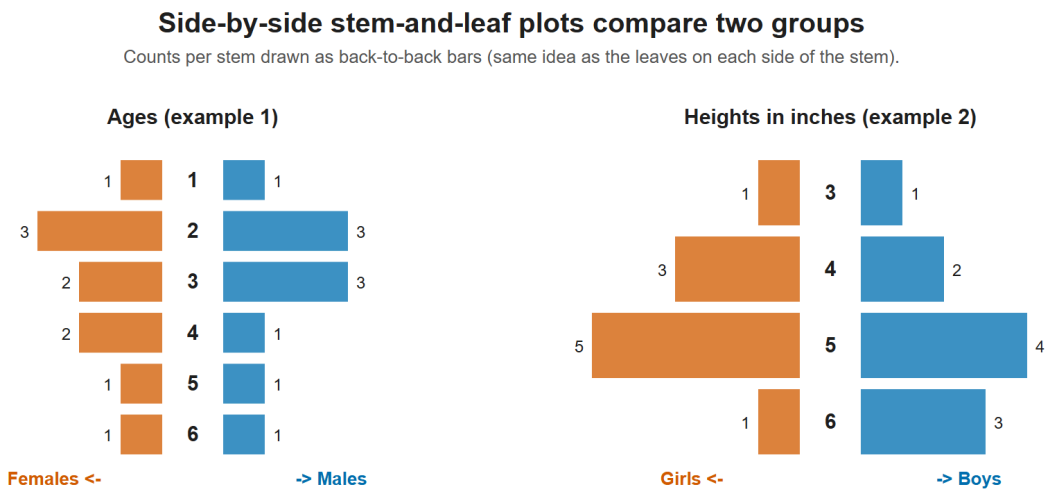
Example 1 - ages of females and males

Females (leaves)	Stem	Males (leaves)
8	1	9
8 7 5	2	2 6 9
5 2	3	5 5 6
7 3	4	6
2	5	8
5	6	3

Key: `8 | 1 |` = 18 (female); `| 1 | 9` = 19 (male).

- Males: 19, 22, 26, 29, 35, 35, 36, 46, 58, 63
- Females: 18, 25, 27, 28, 32, 35, 43, 47, 52, 65

On the female side the smallest leaf sits **next to** the stem. For the 20s, the leaves read `8 7 5`, which means 25, 27, 28.



The number of leaves on each side of each stem, drawn as back-to-back bars. The heights example makes the comparison easy to see: the girls cluster in the 50s while the boys spread further into the 60s.

Example 2 - heights (inches) of girls and boys

Girls (leaves)	Stem	Boys (leaves)
9	3	7
8 6 5	4	7 8
6 4 3 2 1	5	1 6 7 8
0	6	2 3 5

- Girls: 39, 45, 46, 48, 51, 52, 53, 54, 56, 60
- Boys: 37, 47, 48, 51, 56, 57, 58, 62, 63, 65

Comparing the two sides shows that more boys are in the 60s, while the girls are concentrated in the 50s.

Key takeaways

- One shared stem column, with one group's leaves on each side.
- Leaves on the **left** are written in reverse, smallest next to the stem.
- Side-by-side plots make it easy to compare the **center** and **spread** of two groups.

Lesson 2.8 - How to Make a Histogram Using a Frequency Distribution Table

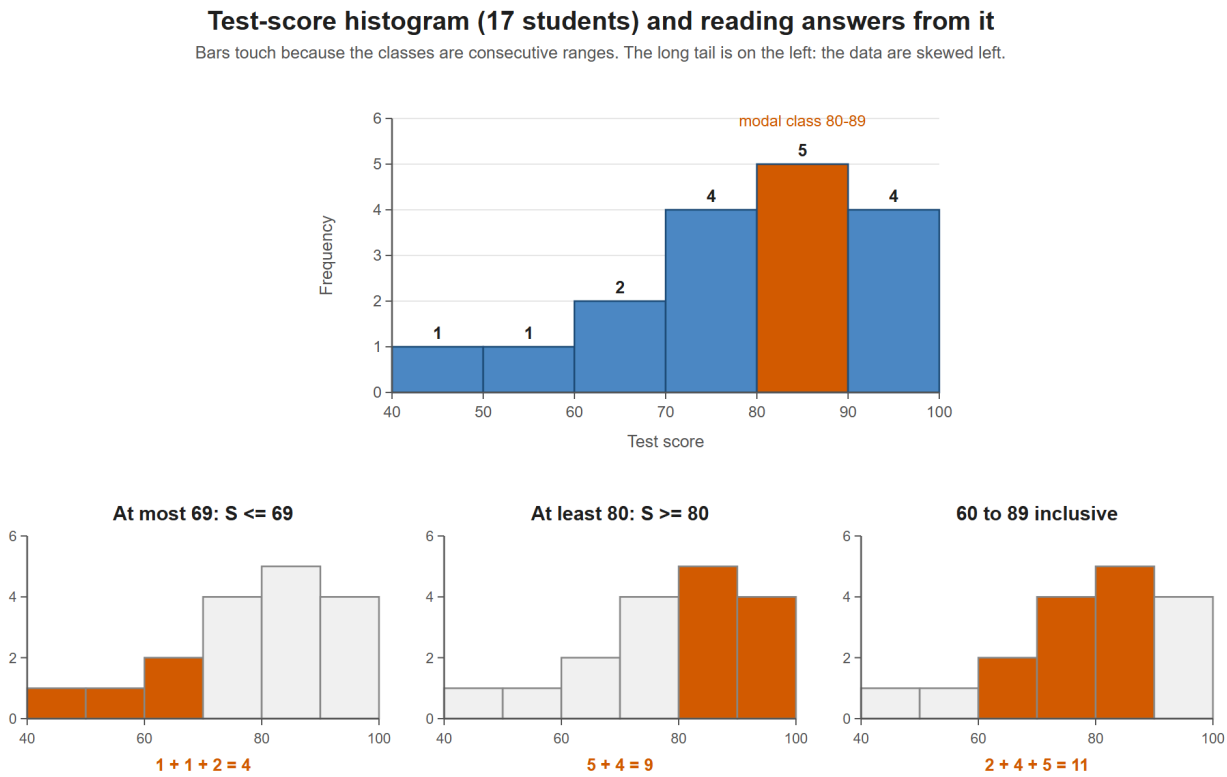
> **Video:** How To Make a Histogram Using a Frequency Distribution Table by The Organic Chemistry Tutor
> **Chapter:** 2 - Graphs and Tables

Steps

1. **Group the data into classes**, ranges of equal width. For test scores, a width of 10 is natural: 40-49, 50-59, ...
2. **Count the frequency** of each class to make a **frequency distribution table**.
3. **Draw the axes:**
   - x-axis (independent variable): the classes, such as the scores.
   - y-axis (dependent variable): the frequency.
4. **Draw a bar for each class**, with height equal to its frequency. The bars **touch**, which is what makes it a histogram rather than a bar graph.

Example - test scores of a class

Score range	Grade	Frequency
40-49	F	1
50-59	F	1
60-69	D	2
70-79	C	4
80-89	B	5
90-100	A	4
Total		**17**



The top chart is the histogram. Its bars touch, and the modal class 80-89 is shown in orange. The three small charts below shade the bars used to answer each question in the next section.

Shape and mode

- The bars are short on the left and tall on the right, so the **long tail is on the left**. The data is **skewed left** (negative skew).
- The **modal class** (the class with the highest frequency) is **80-89**.

Answering questions from a histogram

Let S be a student's score.

1. How many students scored *at most* 69? "At most 69" means $S \leq 69$, so 69 is the maximum.

$$\#(S \leq 69) = \text{sum of the frequencies of every class up to } 69 = 1 + 1 + 2 = 4 \text{ students}$$

2. How many scored *at least* 80? "At least 80" means $S \geq 80$, so 80 is the minimum.

$$\#(S \geq 80) = \text{sum of the frequencies of every class from } 80 \text{ up} = 5 + 4 = 9 \text{ students}$$

3. How many scored between 60 and 89, inclusive? "Inclusive" means 60 and 89 are counted.

$$\#(60 \leq S \leq 89) = \text{sum of the frequencies of the classes from } 60 \text{ to } 89 = 2 + 4 + 5 = 11 \text{ students}$$

> "Between 60 and 89" **exclusive** would mean 61 to 88. Check whether a question says inclusive or exclusive.

Key takeaways

- Build the **frequency distribution table first**, then draw the histogram.
- Histogram bars **touch**; bar-graph bars don't.
- Long tail on the left → skewed left. Long tail on the right → skewed right.
- "At most" = $\leq$, "at least" = $\geq$, "inclusive" = endpoints included.

Lesson 2.9 - How to Make a Frequency Polygon

> **Video:** How To Make a Frequency Polygon by The Organic Chemistry Tutor **Chapter:** 2 - Graphs and Tables

Idea

A **frequency polygon** is a line graph of a frequency distribution. Plot the frequency of each class at the class's **midpoint**, then connect the points with straight lines. Add one extra point **at frequency 0** at each end, so the shape closes onto the x-axis.

Steps

1. Make a **frequency distribution table** with equal-width classes.
2. Find each class **midpoint**: $\text{midpoint} = \dfrac{\text{lower bound} + \text{upper bound}}{2}$
3. Add one **extra midpoint below** the first (first − class width) and one **above** the last (last + class width), each with frequency **0**.
4. Plot (midpoint, frequency) and connect the dots with straight lines.

Example - incomes in a town (class width $20k)

Class boundaries. Each class includes its lower bound but **not** its upper bound, so an income of exactly $40k goes in the 40k-60k class:

$$20 \le I < 40, \quad 40 \le I < 60, \quad \dots$$

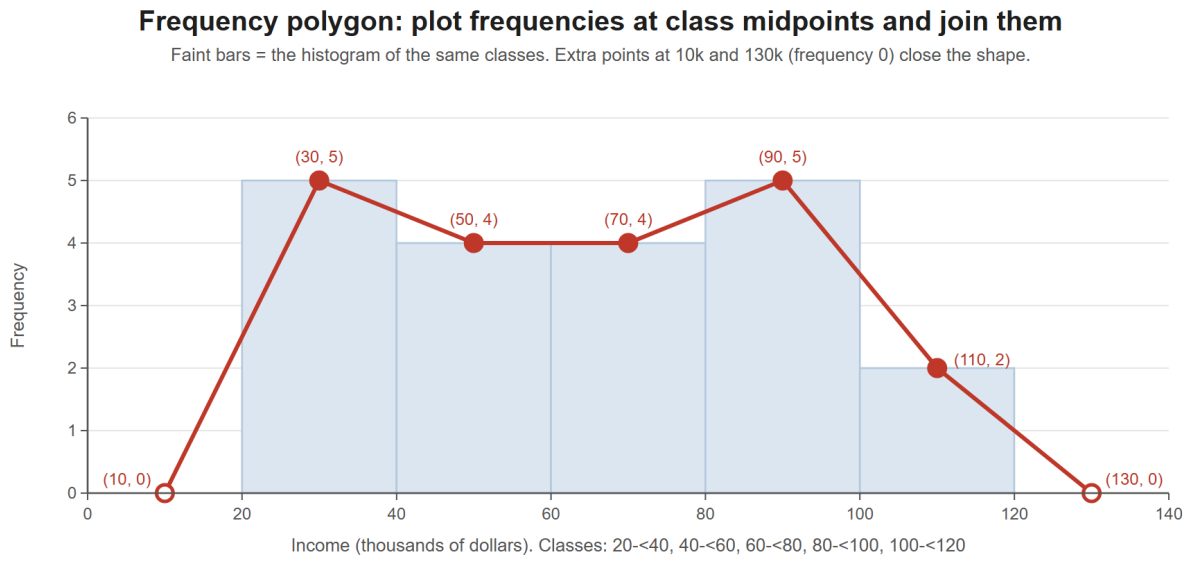
No income is below $20k, and the highest is $115k, so the classes run from 20k to 120k.

Income range (k = $1,000)	Frequency	Midpoint
20k - < 40k	5	30k
40k - < 60k	4	50k
60k - < 80k	4	70k
80k - < 100k	5	90k
100k - < 120k	2	110k
Total	**20**	

For example, the 20k-40k class contains 32k, 25k, 39k, 26k and 25k, so its frequency is 5, and its midpoint is $(20 + 40)/2 = 30$k.

Extra end points: $30 - 20 = 10$k and $110 + 20 = 130$k, both with frequency 0.

Midpoint	10k	30k	50k	70k	90k	110k	130k
Frequency	0	5	4	4	5	2	0



Each point sits at a class **midpoint**, at the height of that class's bar. The two hollow points at 10k and 130k have frequency 0 and bring the polygon down to the axis at both ends.

Key takeaways

- Plot frequencies at **class midpoints**, not at class edges.
- Add one point with **frequency 0** on each side (one class width beyond the first and last midpoints) so the polygon starts and ends on the x-axis.
- State clearly which class a boundary value such as 40k belongs to. A common convention is $[\text{lower}, \text{upper})$.